Building AI Agents

with Spring AI

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A Neutral Home for Agentic AI

The Agentic AI Foundation hosts six open projects under the Linux Foundation

Agentic AI Foundation

"The internet runs on open protocols and open source. So will agentic AI."

PLATINUM MEMBERS

AWS, Anthropic, Block, Bloomberg, Cloudflare, Google, Microsoft, OpenAI

MCP
Connects agents to tools and data

A2A
Lets agents work together

agentgateway
Gateway for agents, models and tools

Agent Skills
AI gateway built on Envoy

goose
Open source AI agent for any LLM

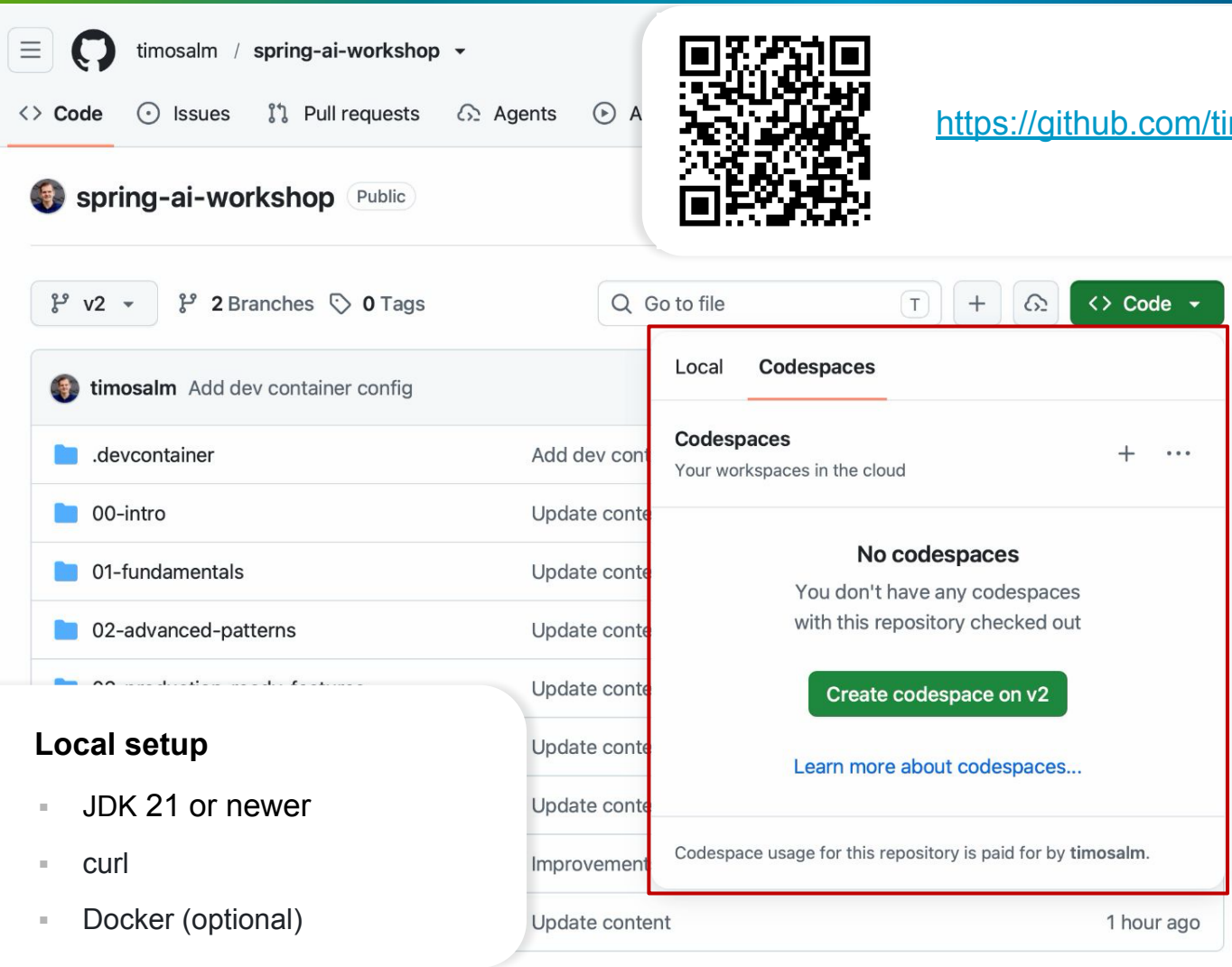
AGENTS.md
Open format to guide coding agents

Highlighted projects come back later in this talk. aaif.io

...

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https://github.com/timosalm/spring-ai-workshop

Local setup

- JDK 21 or newer
- curl
- Docker (optional)

Files

v2

Go to file T

- .devcontainer
- 00-intro
- 01-fundamentals
  - assets
  - sample-app
  - 01-ai-fundamentals.md
  - 02-spring-ai-fundamentals.md
  - 03-advisors.md
  - exercises.md
- 02-advanced-patterns
  - 02-rag
  - 03-tool-calling
    - sample-app
    - 01-tool-calling.md
    - exercises.md

spring-ai-workshop / README.md

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Preview | Code | Blame | 61 lines (43 loc) · 4.2 KB | Raw

Prerequisites

Tooling

- **JDK 21** or newer. The sample apps target `java.version=21`.
- A code editor or IDE, such as IntelliJ IDEA or VS Code.
- **Maven** is not needed separately, because each sample app ships the Maven Wrapper (`./mvnw`).
- **Docker** for the observability lab (Grafana stack) and the MCP security part (Keycloak). Both run through the `compose.yaml` of the app.
- `curl` to call the endpoints, and `jq` for readable JSON output in some labs.

A model provider. The labs use **OpenAI** with the `gpt-5.6-sol` chat model and the `text-embedding-3-small` embedding model. Set your key in the terminal where you run an app.

```
export SPRING_AI_OPENAI_BASE_URL=https://devoxx-be.openai.azure.com
export OPENAI_API_KEY=$(curl -s https://gist.githubusercontent.com/timosalm/a
```

To use another provider, swap the Spring AI starter dependency and the `spring.ai.<provider>.*` properties.

A Model Alone Is Not Enough

The Model

- Makes up facts
- Forgets past conversations
- Has outdated knowledge
- Does not know your data
- Cannot take action

Models are probabilistic, so the same prompt can lead to different answers.

The Usual Fixes Lead to

- Higher token costs
- New attack surfaces
- Governance challenges
- Little observability
- More code to maintain

Retrieval, memory and tools each solve one problem but add new ones.

The Answer

You need a harness around the model.

A harness connects the model to your data, memory and tools. It also keeps every call secure, governed and observable.

Spring AI helps you build this harness.

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Spring AI Fundamentals

One consistent programming model for every AI provider, through two complementary APIs

ChatModel

The low level API behind every provider.

```
ChatResponse response = chatModel.call(
    new Prompt("Tell me about Spring
AI"));

String text = response.getResult()
    .getOutput().getText();
```

Use it when you need explicit control over every request.

ChatClient

The recommended fluent API. It builds on ChatModel and adds useful defaults.

```
String answer = chatClient.prompt()
    .user("Tell me about Spring AI")
    .call()
    .content();
```

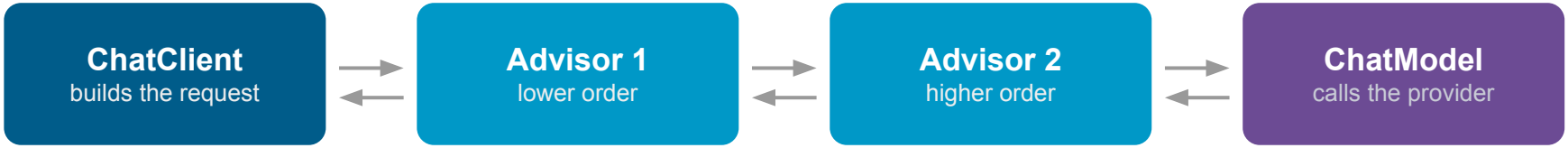
Use it for everyday work in your applications.

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The Advisors API

An advisor is an interceptor around every ChatClient call, with a before and an after step



Each advisor gets a ChatClientRequest on the way in and a ChatClientResponse on the way out.

```
class LoggingAdvisor implements CallAdvisor {
  public ChatClientResponse adviseCall(
      ChatClientRequest req, CallAdvisorChain chain) {
    log(req);                          // before
    var res = chain.nextCall(req);     // rest of chain
    log(res);                          // after
    return res;
  }
  public String getName() { return "LoggingAdvisor"; }
  public int getOrder() { return LOWEST_PRECEDENCE; }
}
```

CallAdvisor and StreamAdvisor

Implement one or both to support the blocking call and the reactive stream.

Name and order

A lower order value runs first on the way in and last on the way out.

Shared context

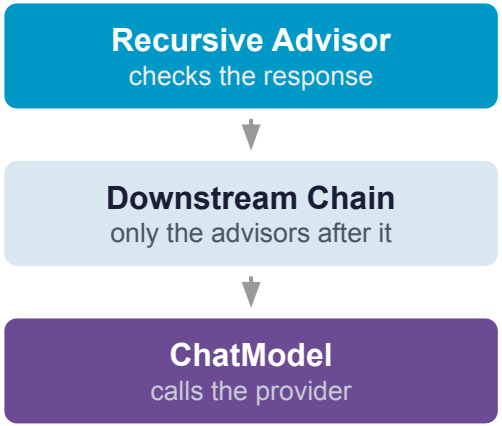
A context map travels with the request, for example to pass a conversation id.

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Recursive Advisors

Some advisors need to call the model more than once for a single user request



Not valid yet?
Try again with feedback.

```
CallAdvisorChain downstream = chain.copy(this);
ChatClientResponse res = downstream.nextCall(req);
int attempt = 1;
while (!matchesSchema(res) && attempt < max) {
  req = addFeedback(req, res);
  res = downstream.nextCall(req);
  attempt++;
}
return res;
```

chain.copy(this) makes sure the advisors before it do not run again.

Examples: ToolCallingAdvisor, StructuredOutputValidationAdvisor

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Hands-on

Spring AI Fundamentals



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Retrieval-Augmented Generation (RAG)

Give the model the right knowledge at the moment it answers



The model itself does not change. RAG only changes what goes into the prompt, which reduces hallucinations.

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RAG in Spring AI

A small set of APIs covers indexing, and retrieval is simply another advisor

Indexing

DocumentReader, DocumentTransformer, and DocumentWriter form the pipeline.

```
var reader = new PagePdfDocumentReader(pdf);
var chunks = TokenTextSplitter.builder().build()
    .apply(reader.get());
vectorStore.add(chunks);
```

A VectorStore is itself a DocumentWriter. Readers exist for PDF, Markdown, JSON, HTML, and more.

Retrieval

QuestionAnswerAdvisor searches the vector store and adds the best matches to the prompt.

```
var rag = QuestionAnswerAdvisor
    .builder(vectorStore)
    .searchRequest( SearchRequest .builder()
        .topK(2).similarityThreshold(0.7)
        .build())
    .build();

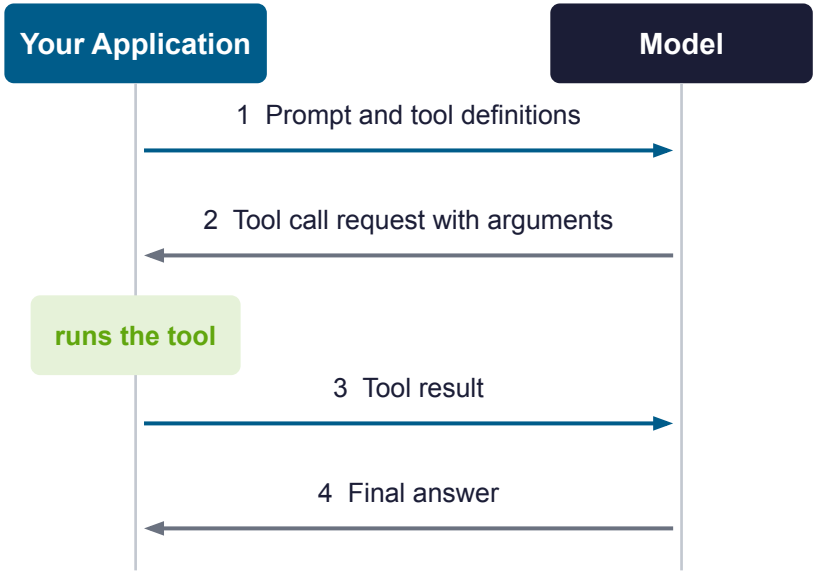
String answer = chatClient.prompt()
    .user( "Which recipes use cheese?" )
    .advisors(rag)
    .call().content();
```

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Tool Calling

The model decides which tool to use. Your application runs it



Tool Calling in Spring AI

```
@Tool(description = "Lists ingredients at home")
List<String> ingredientsAtHome() { ... }

Recipe recipe = chatClient.prompt()
    .user("Suggest a recipe with cheese")
    .tools(this)
    .call().entity(Recipe.class);
```

Annotations become definitions Spring AI turns each @Tool method into a name, a description, and an input schema.

The loop is handled for you Spring AI runs requested tools and sends the results back until the answer is ready.

Secure by design API keys and data access stay in your app.

Tools let the model act, and that is the first step from a chatbot to an agent.

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Hands-on

Advanced Patterns



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Production-Ready Features

Spring AI plugs into the Spring tools you already use for observability and testing

Observability

Built on Spring Boot

Micrometer and Actuator send metrics and traces to Prometheus, Zipkin, or OpenTelemetry.

Instrumented out of the box

ChatClient, advisors, chat models, tool calls, embedding models, image models, and vector stores.

Token usage as a metric

Input and output tokens are counted per call, so you can track cost.

Content logging is opt-in

Prompts, completions, and tool arguments stay out of logs unless you enable them.

Testing

Split your tests by the kind of code, and run real services in Docker when you need them.

Deterministic shell

Mock the ChatModel and test prompts, tools, and mapping with JUnit.

Evaluators

RelevancyEvaluator and FactCheckingEvaluator let a model judge the answer.

Testcontainers `spring-ai-spring-boot-testcontainers`

@ServiceConnection connects your app automatically to Ollama, Chroma, Qdrant, Milvus, Weaviate, and the Docker MCP Gateway.

```
@Container @ServiceConnection
static OllamaContainer ollama =
    new OllamaContainer("ollama/ollama");
```

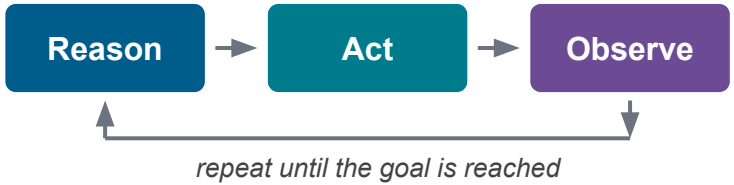
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What Is Agentic AI?

From answering single questions to reaching goals on its own

An AI agent uses a model to decide its next step, acts through tools, and repeats this loop until the goal is reached.



Agentic systems come in two forms. A workflow follows a path you define in code, while an agent chooses its own path.

THE EVOLUTION OF LLM BASED SOLUTIONS

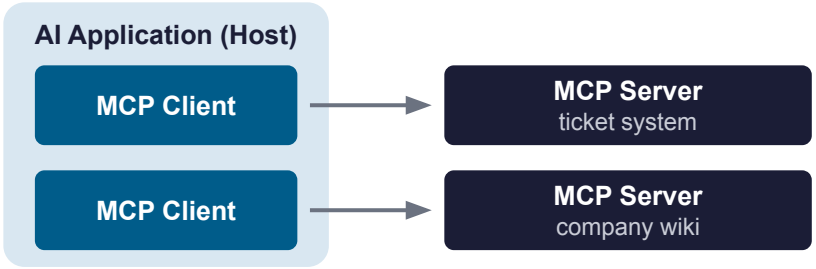
- **Standalone LLMs**
  Prompts in, text out for language tasks
- **Grounded Generation (RAG)**
  Answers based on your own business data
- **Tool Calling and MCP**
  The model can trigger actions in real systems
- **Agentic Systems**
  Workflows and agents built on models and tools

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Excursion: Model Context Protocol (MCP)

An open standard that connects AI applications to tools and data, like USB-C for AI

What MCP Is



Three primitives

Tools to act, resources for context, and reusable prompts.

Two transports

stdio locally, Streamable HTTP over the network.

Open and vendor neutral

Governed by the Agentic AI Foundation, not one vendor.

MCP in Spring AI

As a client

`spring-ai-starter-mcp-client`

```
@Bean
ChatClient chatClient(ChatClient.Builder cb,
    ToolCallbackProvider mcpTools) {
  return cb.defaultTools(mcpTools).build();
}
```

As a server

`spring-ai-starter-mcp-server-webmvc`

```
@McpTool(description = "Get project releases")
List<Release> releases(
    @McpToolParam(description = "Project slug")
    String slug) { ... }
```

Remote MCP tools behave like your own @Tool methods. The same tool calling loop runs, and your prompting code stays the same.

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Hands-on

Model Context Protocol



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Two Approaches to Orchestrate LLMs and Tools

Workflows

Your code defines the path, and the model does its work inside each step.

- ✓ Predictable and consistent results
- ✓ Easier to test and debug
- ✓ Lower cost and latency
- ✓ Ideal for well defined, repeatable tasks
- ✗ Cannot adapt to unexpected subtasks
- ✗ You must split the task up front

Pure Agents

The model decides which steps to take and which tools to use.

- ✓ Flexible for open ended problems
- ✓ Handles tasks whose steps are unknown up front
- ✓ Scales well in trusted environments
- ✗ Higher cost and latency
- ✗ Errors can add up over many steps
- ✗ Harder to debug and control
- ✗ Needs thorough testing and guardrails

Start with a workflow for predictability. Use an agent only when you truly need its flexibility.

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Workflow Patterns

Your code decides the steps. Pick the simplest pattern that fits the task

Chain

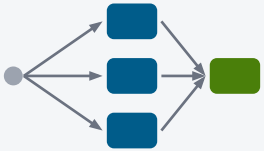


Splits a task into steps. Each output feeds the next model call.

Sample code

`spring-ai-examples`

Parallelization

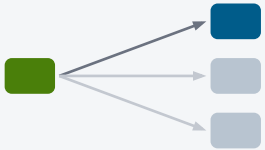


Runs independent calls at the same time and combines the results.

Sample code

`spring-ai-examples`

Routing

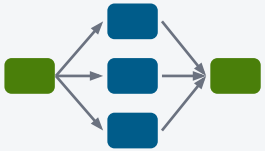


Classifies the input and sends it to a specialized prompt or model.

Sample code

`spring-ai-examples`

Orchestrator-Workers



Splits the task at runtime and lets workers handle the parts.

Sample code

`spring-ai-examples`

Evaluator-Optimizer



Generates an answer, evaluates it, and retries with the feedback.

Experimental

`SelfRefineEvaluationAdvisor`

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Patterns for a Single Agent

Components in the harness that keep one agent focused, on track, and under control

Agent Skills

Reusable instruction packages. Only the name and description are loaded at the start, and the full instructions follow when a task needs them.

Experimental `SkillsTool`

Tool Search

The agent gets one search tool and loads only the few tool definitions it needs for the current step.

GA `ToolSearchToolCallingAdvisor`

Plan and Execute

The agent writes a task list first and works through it step by step, so it does not lose track.

Experimental `TodoWriteTool`

Human in the Loop

The agent asks when the input is unclear and asks for approval before risky actions.

Experimental `AskUserQuestionTool`

Tool Search is part of Spring AI since 2.0. The other tools come from the community project spring-ai-agent-utils.

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Multi-Agent Patterns

Keep one agent in charge, and add a second one only where it pays off

LLM as a Judge

A second agent rates the answer and sends feedback until it passes. A different judge model avoids bias.

Experimental

`SelfRefineEvaluationAdvisor`

Subagents

The main agent delegates a task to an agent with its own context, tools, and model, and reads only the result.

Experimental

`TaskTool`

Agent-to-Agent (A2A)

An open protocol that lets agents in different systems find and call each other, like MCP does for tools.

Experimental

`spring-ai-a2a`

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Hands-on

Agentic AI

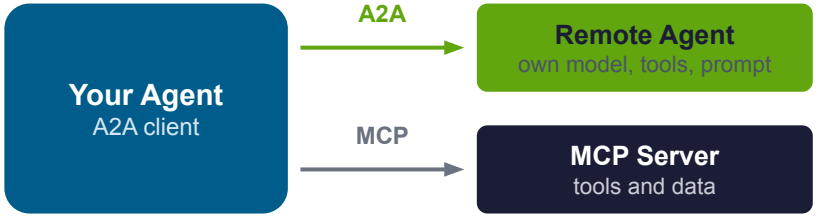


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Excursion to the Agent-to-Agent Protocol (A2A)

MCP connects an agent to tools. A2A connects an agent to other agents

What A2A Is



Agent Card

A JSON description of the agent and its skills, published at a well known URL.

Messages and tasks

Agents talk over JSON-RPC and track longer work as tasks.

Open and vendor neutral

Started by Google and now hosted by the Linux Foundation.

A2A in Spring AI

Experimental

As a server `spring-ai-a2a-server-autoconfigure`

1. **Your capabilities** Still defined as usual, with @Tool methods, advisors, and prompts on your ChatClient.
2. **An AgentCard bean** Describes the name, description, and skills of your agent for discovery.
3. **An AgentExecutor bean** Hands incoming messages to your ChatClient, usually via DefaultAgentExecutor.

Spring AI publishes the card and handles the protocol for you.

As a client `a2a-java-sdk-client`

Wrap the call to a remote agent in a @Tool method. The model then decides on its own when to delegate.

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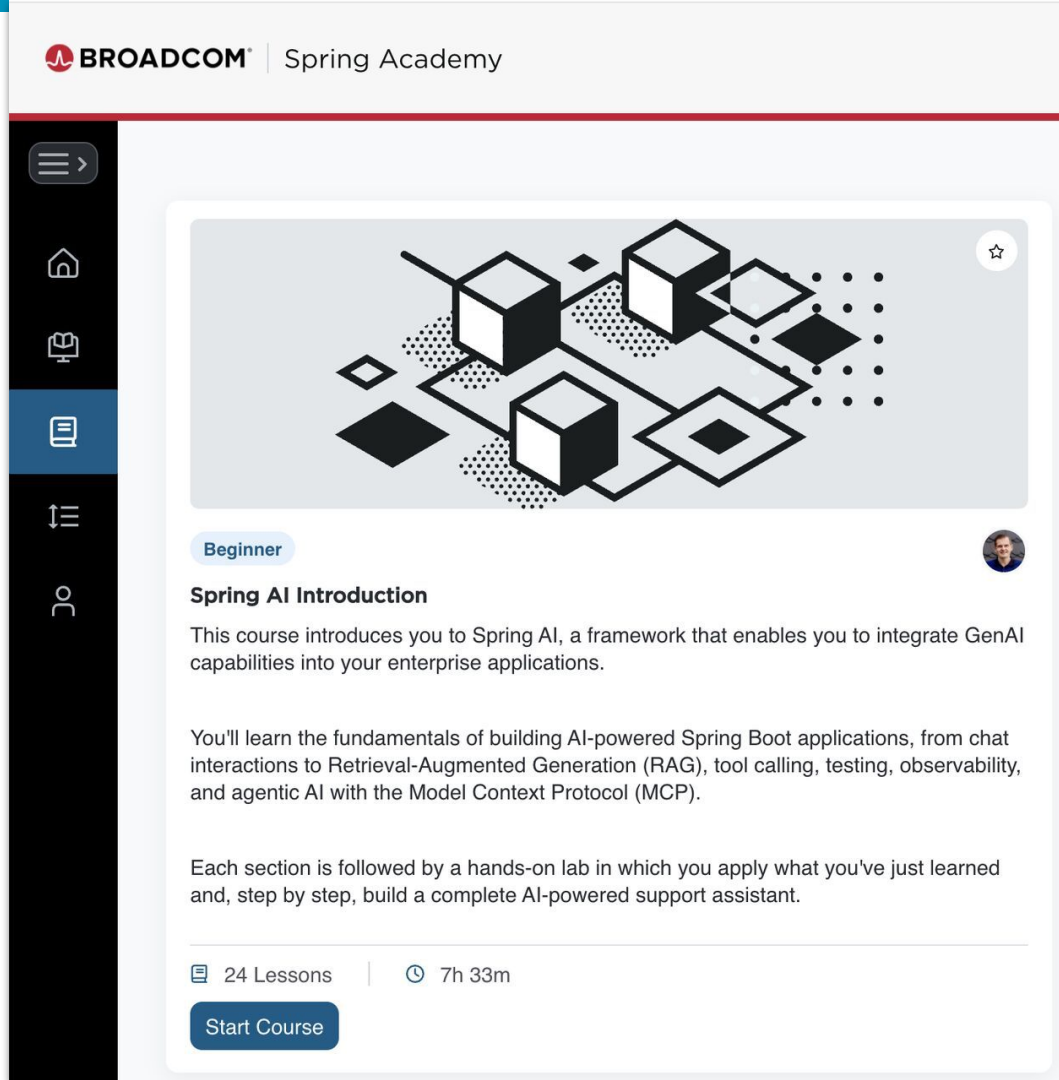
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https://github.com/timosalm/spring-ai-workshop

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